Secure Certificate Management System

Software Requirements Specification

IEEE 830-1998 outline

DRAFT — human review required

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1 Introduction

1.1 Purpose

This specification records the project purpose, stakeholder constraints and unresolved decisions stated in the source evidence below.

This SRS describes the software's required behavior for users, administrators and the Project Analyst reviewing the specification.

1.2 Scope

Design a secure digital-certificate management system for a university that manages certificates used by approved campus applications, servers, and network services. Authorized certificate administrators should be able to review certificate requests, approve certificate issuance, renew certificates, revoke compromised certificates, and track certificate status. The system should restrict certificate administration based on approved roles, monitor certificate validity, detect expired certificates and unauthorized certificate changes, generate security alerts, and maintain audit logs of certificate lifecycle activities and administrator actions. The platform should provide a centralized dashboard, integrate with existing university identity services, support multiple campus locations, and remain secure, reliable, scalable, and manageable.

1.3 Definitions, acronyms and abbreviations

Term	Meaning
Log	A collection of event records. This term does not itself require additional logging fields, events or access permissions.
Audit record	A stored record of an event. Required event scope and retention are determined by the linked source, not this definition.

1.4 References

Reference	Source
Format	IEEE Std 830-1998, clauses 5 and Annex A (document organization).

1.5 Overview

Section 2 summarizes the product, stakeholders and open decisions. Section 3 contains the uniquely identified requirements. Detailed evidence, acceptance procedures are available in the Normal report. Appendices A-C contain derived threat hypotheses, risk assessment and the proposed SDLC workflow.

2 Overall description

2.1 Product perspective

Design a secure digital-certificate management system for a university that manages certificates used by approved campus applications, servers, and network services.

2.2 Product functions

Functional topics are summarized below; section 3 retains the full requirements.

- manage certificates used by approved campus applications, servers, and network services - FR-001
- allow Authorized certificate administrators to review certificate requests - FR-002
- allow Authorized certificate administrators to approve certificate issuance - FR-003
- allow Authorized certificate administrators to renew certificates - FR-004
- allow Authorized certificate administrators to revoke compromised certificates - FR-005
- allow Authorized certificate administrators to track certificate status - FR-006
- restrict certificate administration based on approved roles - FR-007
- monitor certificate validity - FR-008
- detect expired certificates - FR-009
- detect unauthorized certificate changes - FR-010
- generate security alerts - FR-011
- maintain audit logs of certificate lifecycle activities and administrator actions - FR-012
- provide a centralized dashboard - FR-013
- integrate with existing university identity services - FR-014
- support multiple campus locations - FR-015

2.3 User characteristics and stakeholders

Labels group stakeholders for readability. They do not combine permissions.

Education, training and experience requirements remain unspecified.

Role	Source user groups	Responsibility
Admin	administrators	Performs only the role's source-defined operations in section 3.
Project Analyst	Document review role	Reviews requirements and tracks open decisions; this role grants no product access.

2.4 Constraints

No additional implementation constraints were confirmed. Source-defined restrictions remain in the requirements in section 3.

2.5 Assumptions and dependencies

Open decisions requiring confirmation:

- What conditions define the requested detections and their alert triggers?
- Which fields must the audit logs contain?
- What retention period is required for the audit logs?
- Which security acceptance outcomes must be demonstrated?
- What measurable reliability and recovery targets must the system meet?
- What measurable scalability targets must the system meet?
- What criteria define acceptable manageability?
- How are the stated roles and authorized permissions assigned?

3 Specific requirements

Each requirement is stated once with its original identifier and wording. Cross-references identify requirements covering more than one concern.

3.1 External interface requirements

3.1.1 User interfaces

Related requirements: FR-013. Interface details beyond those requirements remain unspecified.

3.1.2 Hardware interfaces

No specific interface requirement was supplied; confirmation is required.

3.1.3 Software interfaces

Related requirements: FR-014. Interface details beyond those requirements remain unspecified.

3.1.4 Communications interfaces

Related requirements: FR-001. Interface details beyond those requirements remain unspecified.

3.2 Functional requirements

FR-001 · Must · explicit

The system shall manage certificates used by approved campus applications, servers, and network services.

Traceability: E-001, TEST-001

FR-002 · Should · explicit

The system shall allow Authorized certificate administrators to review certificate requests.

Traceability: E-002, TEST-002

FR-003 · Should · explicit

The system shall allow Authorized certificate administrators to approve certificate issuance.

Traceability: E-002, TEST-003

FR-004 · Should · explicit

The system shall allow Authorized certificate administrators to renew certificates.

Traceability: E-002, TEST-004

FR-005 · Should · explicit

The system shall allow Authorized certificate administrators to revoke compromised certificates.

Traceability: E-002, TEST-005

FR-006 · Should · explicit

The system shall allow Authorized certificate administrators to track certificate status.

Traceability: E-002, TEST-006

FR-007 · Should · explicit

The system shall restrict certificate administration based on approved roles.

Traceability: E-003, TEST-007

FR-008 · Should · explicit

The system shall monitor certificate validity.

Traceability: E-003, TEST-008

FR-009 · Should · explicit

The system shall detect expired certificates.

Traceability: E-003, TEST-009

FR-010 · Should · explicit

The system shall detect unauthorized certificate changes.

Traceability: E-003, TEST-010

FR-011 · Should · explicit

The system shall generate security alerts.

Traceability: E-003, TEST-011

FR-012 · Should · explicit

The system shall maintain audit logs of certificate lifecycle activities and administrator actions.

Traceability: E-003, TEST-012

FR-013 · Should · explicit

The system shall provide a centralized dashboard.

Traceability: E-004, TEST-013

FR-014 · Should · explicit

The system shall integrate with existing university identity services.

Traceability: E-004, TEST-014

FR-015 · Should · explicit

The system shall support multiple campus locations.

Traceability: E-004, TEST-015

3.3 Performance requirements

NFR-004 · Should · explicit

The system shall remain scalable.

Traceability: E-004, TEST-019

3.4 Logical database requirements

No separate requirement was specified in the supplied brief.

3.5 Design constraints

No separate requirement was specified in the supplied brief.

3.6 Software system attributes

3.6.1 Reliability

NFR-003 · Should · explicit

The system shall remain reliable.

Traceability: E-004, TEST-018

3.6.2 Availability

No separate requirement was specified in the supplied brief.

3.6.3 Security

Also covered by: FR-007, FR-011, FR-012.

NFR-001 · Must · explicit

The system shall be secure.

Traceability: E-001, TEST-016

NFR-002 · Should · explicit

The system shall remain secure.

Traceability: E-004, TEST-017

3.6.4 Maintainability

NFR-005 · Should · explicit

The system shall remain manageable.

Traceability: E-004, TEST-020

3.6.5 Portability

No separate requirement was specified in the supplied brief.

3.7 Other requirements

No separate requirement was specified in the supplied brief.

A Derived threat hypotheses

These are defensive planning hypotheses, not observed incidents or approved extra controls. Unrecognized threats and actual implementation effectiveness require human review.

These are planning hypotheses, not observed incidents or additional approved controls.

A.1 THR-001: User identities and authenticated access

Scenario: Derived hypothesis: An attacker uses a compromised identity or authenticated session to impersonate an authorized user and exercise that user's permitted operations.

Hypothetical actor or cause: Hypothetical attacker, privileged insider or failure cause

Potential consequence: Unauthorized use of a legitimate identity could expose resources or alter system state.

Linked requirements: FR-013, FR-014, FR-015, NFR-002, NFR-003, NFR-004, NFR-005

Original evidence: E-004

Residual exposure: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

A.2 THR-002: Role and permission boundaries

Scenario: Derived hypothesis: A principal exploits an incorrect role assignment or permission check to perform operations outside the source-defined authorization boundary.

Hypothetical actor or cause: Hypothetical attacker, privileged insider or failure cause

Potential consequence: A lower-privilege user could access restricted functions or sensitive resources.

Linked requirements: FR-002, FR-003, FR-004, FR-005, FR-006, FR-007, FR-008, FR-009, FR-010, FR-011, FR-012

Original evidence: E-002, E-003

Residual exposure: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

A.3 THR-003: Access grants and revocations

Scenario: Derived hypothesis: A revoked or changed permission remains effective in a session or enforcement point, allowing access after authorization should have ended.

Hypothetical actor or cause: Hypothetical attacker, privileged insider or failure cause

Potential consequence: A formerly authorized principal could continue using access that is no longer approved.

Linked requirements: FR-002, FR-003, FR-004, FR-005, FR-006

Original evidence: E-002

Residual exposure: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

A.4 THR-004: Integration trust paths

Scenario: Derived hypothesis: A compromised connected system or integration path supplies forged, altered or missing identity, event or policy data used by the source-defined integration.

Hypothetical actor or cause: Hypothetical attacker, privileged insider or failure cause

Potential consequence: Untrusted integration data could cause incorrect decisions or hide security events.

Linked requirements: FR-013, FR-014, FR-015, NFR-002, NFR-003, NFR-004, NFR-005

Original evidence: E-004

Residual exposure: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

A.5 THR-005: Security and administrative audit evidence

Scenario: Derived hypothesis: An attacker or recording failure suppresses or alters source-required audit evidence so that access decisions, security events or admin actions cannot be reconstructed.

Hypothetical actor or cause: Hypothetical attacker, privileged insider or failure cause

Potential consequence: Investigators could lose reliable evidence of the events required by the brief.

Linked requirements: FR-007, FR-008, FR-009, FR-010, FR-011, FR-012

Original evidence: E-003

Residual exposure: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

A.6 THR-006: Monitoring and security alerts

Scenario: Derived hypothesis: Suspicious activity evades the source-defined detection workflow, or excessive events obscure actionable alerts before an authorized reviewer can investigate.

Hypothetical actor or cause: Hypothetical attacker, privileged insider or failure cause

Potential consequence: Relevant suspicious activity could remain undetected or receive delayed investigation.

Linked requirements: FR-007, FR-008, FR-009, FR-010, FR-011, FR-012

Original evidence: E-003

Residual exposure: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

B Risk assessment

Risk treatments are proposals for review. Unknown likelihood and impact ratings mean unassessed; no numeric confidence or severity is inferred.

B.1 RISK-001

Derived risk affecting User identities and authenticated access: Unauthorized use of a legitimate identity could expose resources or alter system state.

Likelihood: unknown; impact: unknown; owner: TBD.

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: An attacker uses a compromised identity or authenticated session to impersonate an authorized user and exercise that user's permitted operations.

Proposed treatment: Review impersonation and compromised-identity paths against the stated access duties; confirm any additional identity or session protections with the stakeholder. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Linked requirements: FR-013, FR-014, FR-015, NFR-002, NFR-003, NFR-004, NFR-005

Original evidence: E-004

B.2 RISK-002

Derived risk affecting Role and permission boundaries: A lower-privilege user could access restricted functions or sensitive resources.

Likelihood: unknown; impact: unknown; owner: TBD.

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: A principal exploits an incorrect role assignment or permission check to perform operations outside the source-defined authorization boundary.

Proposed treatment: Test permitted and denied role/resource pairs against the linked authorization duties; review ambiguous permission assignments with the resource owner. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Linked requirements: FR-002, FR-003, FR-004, FR-005, FR-006, FR-007, FR-008, FR-009, FR-010, FR-011, FR-012

Original evidence: E-002, E-003

B.3 RISK-003

Derived risk affecting Access grants and revocations: A formerly authorized principal could continue using access that is no longer approved.

Likelihood: unknown; impact: unknown; owner: TBD.

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: A revoked or changed permission remains effective in a session or enforcement point, allowing access after authorization should have ended.

Proposed treatment: Trace permission changes through subsequent access decisions and any active access; confirm revocation timing and session semantics before baselining them. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Linked requirements: FR-002, FR-003, FR-004, FR-005, FR-006

Original evidence: E-002

B.4 RISK-004

Derived risk affecting Integration trust paths: Untrusted integration data could cause incorrect decisions or hide security events.

Likelihood: unknown; impact: unknown; owner: TBD.

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: A compromised connected system or integration path supplies forged, altered or missing identity, event or policy data used by the source-defined integration.

Proposed treatment: Review trust assumptions at the specified integration boundaries with missing or altered input fixtures; confirm the required failure handling with the owner. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Linked requirements: FR-013, FR-014, FR-015, NFR-002, NFR-003, NFR-004, NFR-005

Original evidence: E-004

B.5 RISK-005

Derived risk affecting Security and administrative audit evidence: Investigators could lose reliable evidence of the events required by the brief.

Likelihood: unknown; impact: unknown; owner: TBD.

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: An attacker or recording failure suppresses or alters source-required audit evidence so that access decisions, security events or admin actions cannot be reconstructed.

Proposed treatment: Compare source-required events with the resulting audit evidence; assess missing and altered-record cases and confirm unprovided protection or retention decisions. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Linked requirements: FR-007, FR-008, FR-009, FR-010, FR-011, FR-012

Original evidence: E-003

B.6 RISK-006

Derived risk affecting Monitoring and security alerts: Relevant suspicious activity could remain undetected or receive delayed investigation.

Likelihood: unknown; impact: unknown; owner: TBD.

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: Suspicious activity evades the source-defined detection workflow, or excessive events obscure actionable alerts before an authorized reviewer can investigate.

Proposed treatment: Evaluate the stated detection duties with representative positive and negative events; confirm undecided thresholds, alert priorities and investigation responsibilities. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Linked requirements: FR-007, FR-008, FR-009, FR-010, FR-011, FR-012

Original evidence: E-003

C SDLC recommendation and delivery workflow

Proposed lifecycle: Incremental

The Incremental model is selected based on derived evidence showing the need for integrated security review (E-002), operational validation through monitoring feedback (E-003), staged deployment across multiple locations (E-004) to enable controlled rollout, and integration verification of connected systems (E-004). These factors collectively support a phased, iterative approach that aligns with continuous security assurance and operational feedback. The Incremental model scores 66 points—outperforming Hybrid (65), DevSecOps (63), Spiral (51), and others—due to superior alignment across derived validation criteria. While the evidence is derived rather than direct, it establishes a consistent need for iterative validation, security integration, and deployment control. The model does not require assumptions about budget, delivery, failure consequences, or legacy systems; these remain unknown factors as specified. The decision remains proposed, pending

confirmation of team capacity and delivery constraints. No product-based suitability is inferred beyond the evidence provided.

Recommendation requires human confirmation of delivery constraints and team capacity. Suitability points are deterministic rule scores, not probabilities.

Policy version: sequential-sdlc-2.0

Unconfirmed decision factors: budget, delivery, failure consequences, feedback, legacy, regulation, schedule, stability, uncertainty, verification

- Recurring delivery requires stakeholder review and backlog refinement; a cycle does not guarantee completeness or remove unresolved decisions.
- Proposed suitability recommendation; confirm delivery constraints and team capacity before adopting this lifecycle. Missing factors are not assumed.

C.1 Ranking and source evidence

- security_assurance = integrated security review (derived). Security decisions benefit from security review and tests throughout delivery. Original evidence: E-002.
- operational_validation = monitoring feedback (derived). Monitoring and alert behavior needs validation with operational feedback; this does not establish continuous deployment or a release frequency. Original evidence: E-003.
- deployment_scope = staged rollout opportunity (derived). Multiple deployment locations offer a staged validation and rollout opportunity; the rollout plan remains proposed. Original evidence: E-004.
- integration_assurance = integration validation (derived). Connected systems need integration verification before broader rollout; their age, technology and technical uncertainty remain unspecified. Original evidence: E-004.

Candidate	Suitability points
Incremental	66
Hybrid	65
DevSecOps	63
Spiral	51
Agile	40
V-Model	40
Waterfall	40

C.1.1 Incremental suitability reasons

- Derived operational_validation (+6 points): Monitoring and alert behavior needs validation with operational feedback; this does not establish continuous deployment or a release frequency. (E-003)
- Derived deployment_scope (+12 points): Multiple deployment locations offer a staged validation and rollout opportunity; the rollout plan remains proposed. (E-004)
- Derived integration_assurance (+8 points): Connected systems need integration verification before broader rollout; their age, technology and technical uncertainty remain unspecified. (E-004)

C.1.2 Hybrid suitability reasons

- Derived security_assurance (+8 points): Security decisions benefit from security review and tests throughout delivery. (E-002)
- Derived operational_validation (+5 points): Monitoring and alert behavior needs validation with operational feedback; this does not establish continuous deployment or a release frequency. (E-003)
- Derived deployment_scope (+8 points): Multiple deployment locations offer a staged validation and rollout opportunity; the rollout plan remains proposed. (E-004)
- Derived integration_assurance (+4 points): Connected systems need integration verification before broader rollout; their age, technology and technical uncertainty remain unspecified. (E-004)

C.1.3 DevSecOps suitability reasons

- Derived security_assurance (+12 points): Security decisions benefit from security review and tests throughout delivery. (E-002)
- Derived operational_validation (+8 points): Monitoring and alert behavior needs validation with operational feedback; this does not establish continuous deployment or a release frequency. (E-003)
- Derived deployment_scope (+3 points): Multiple deployment locations offer a staged validation and rollout opportunity; the rollout plan remains proposed. (E-004)

C.1.4 Spiral suitability reasons

- Derived security_assurance (+5 points): Security decisions benefit from security review and tests throughout delivery. (E-002)
- Derived integration_assurance (+6 points): Connected systems need integration verification before broader rollout; their age, technology and technical uncertainty remain unspecified. (E-004)

C.1.5 Agile suitability reasons

- No differentiating input evidence

C.1.6 V-Model suitability reasons

- No differentiating input evidence

C.1.7 Waterfall suitability reasons

- No differentiating input evidence

C.2 Methodology phases

Activities reference the canonical requirements and proposed verification work. Human gates are pending decisions, not approval records.

C.2.1 Requirements and verification planning

Proposed roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Linked requirements: FR-001, FR-002, FR-003, FR-004, FR-005, FR-006, FR-007, FR-008, FR-009, FR-010, FR-011, FR-012, FR-013, FR-014, FR-015, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005

Human gate: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

- Activity: Review and baseline the source-defined work group: FR-001: The system shall manage certificates used by approved campus applications, servers, and network services.; FR-002: The system shall allow Authorized certificate administrators to review certificate requests.; FR-003: The system shall allow Authorized certificate administrators to approve certificate issuance.; FR-004: The system shall allow Authorized certificate administrators to renew certificates.; FR-005: The system shall allow Authorized certificate administrators to revoke compromised certificates.; FR-006: The system shall allow Authorized certificate administrators to track certificate status.; FR-007: The system shall restrict certificate administration based on approved roles.; FR-008: The system shall monitor certificate validity.; FR-009: The system shall detect expired certificates.; FR-010: The system shall detect unauthorized certificate changes.; FR-011: The system shall generate security alerts.; FR-012: The system shall maintain audit logs of certificate lifecycle activities and administrator actions.; FR-013: The system shall provide a centralized dashboard.; FR-014: The system shall integrate with existing university identity services.; FR-015: The system shall support multiple campus locations.; NFR-001: The system shall be secure.; NFR-002: The system shall remain secure.; NFR-003: The system shall remain reliable.; NFR-004: The system shall remain scalable.; NFR-005: The system shall remain manageable.
- Activity: Create the unit, integration, system and acceptance test plan before implementation; retain each authored acceptance criterion.
- Deliverable: Versioned requirements baseline
- Deliverable: Unit, integration, system and acceptance test plan
- Deliverable: Unresolved-decision record
- Entry criterion: Authoritative inputs and canonical requirements are available for review.
- Exit criterion: A human reviewer has recorded the baseline decision and disposition of unresolved scope.
- Security check: Review implementation and acceptance evidence against FR-011: The system shall generate security alerts.
- Security check: Review implementation and acceptance evidence against NFR-001: The system shall be secure.
- Security check: Review implementation and acceptance evidence against NFR-002: The system shall remain secure.

C.2.2 Incremental delivery cycle

Proposed roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Linked requirements: FR-001, FR-002, FR-003, FR-004, FR-005, FR-006, FR-007, FR-008, FR-009, FR-010, FR-011, FR-012, FR-013, FR-014, FR-015, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005

Human gate: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

- Activity: Review and prioritize the backlog for the next iteration without changing source-defined scope.
- Activity: Implement the source-defined work group: FR-001: The system shall manage certificates used by approved campus applications, servers, and network services.; FR-002: The system shall allow Authorized certificate administrators to review certificate requests.; FR-003: The system shall allow Authorized certificate administrators to approve certificate issuance.; FR-004: The system shall allow Authorized certificate administrators to renew certificates.; FR-005: The system shall allow Authorized certificate administrators to revoke compromised certificates.; FR-006: The system shall allow Authorized certificate administrators to track certificate status.; FR-007: The system shall restrict certificate administration based on approved roles.; FR-008: The system shall monitor certificate validity.; FR-009: The system shall detect expired certificates.; FR-010: The system shall

detect unauthorized certificate changes.; FR-011: The system shall generate security alerts.; FR-012: The system shall maintain audit logs of certificate lifecycle activities and administrator actions.; FR-013: The system shall provide a centralized dashboard.; FR-014: The system shall integrate with existing university identity services.; FR-015: The system shall support multiple campus locations.; NFR-001: The system shall be secure.; NFR-002: The system shall remain secure.; NFR-003: The system shall remain reliable.; NFR-004: The system shall remain scalable.; NFR-005: The system shall remain manageable.

- Activity: In an authorized isolated test environment, execute TEST-001: Given: system manages certificates used by approved campus applications, servers, and network services When: the system initiates certificate management for an approved service Expected: the resulting certificate state is returned in a response captured externally
- Activity: In an authorized isolated test environment, execute TEST-002: Given: Authorized certificate administrators review a certificate request When: the actor initiates the review operation on the named request Expected: the requested certificate request is returned for examination
- Activity: In an authorized isolated test environment, execute TEST-003: Given: Authorized certificate administrators approve a certificate issuance request When: the actor initiates the approval operation on the named request Expected: the request's approved status is observed in the system state
- Activity: In an authorized isolated test environment, execute TEST-004: Given: Authorized certificate administrators renew a certificate When: the actor initiates the renewal operation on the named certificate Expected: the certificate is observed in its renewed state
- Activity: In an authorized isolated test environment, execute TEST-005: Given: Authorized certificate administrators revoke a compromised certificate When: the actor initiates the revocation operation on the named certificate Expected: the certificate's revoked status is observed in the system state
- Activity: In an authorized isolated test environment, execute TEST-006: Given: Authorized certificate administrators track a certificate status When: the actor initiates the tracking operation on the named certificate Expected: the history of that certificate's status is available for examination
- Activity: In an authorized isolated test environment, execute TEST-007: Given: the system has certificate administration privileges assigned to approved roles When: the system attempts to perform certificate administration actions Expected: only actions permitted by the approved role are executed; unauthorized actions are denied
- Activity: In an authorized isolated test environment, execute TEST-008: Given: a certificate is issued with a defined validity period When: the system monitors certificate validity during operation Expected: The system reports that certificate validity is within the defined period during operation
- Activity: In an authorized isolated test environment, execute TEST-009: Given: a certificate has expired or is nearing expiration When: the system detects certificate validity status changes Expected: the system identifies and flags expired certificates within the monitoring scope
- Activity: In an authorized isolated test environment, execute TEST-010: Given: a certificate is modified by an unauthorized entity When: An unauthorized modification to the certificate is detected. Expected: The system generates an alert indicating the change and its source.
- Activity: In an authorized isolated test environment, execute TEST-011: Given: a security condition is triggered that requires alerting When: the system initiates an alert condition Expected: an alert exists in the system as a record, with no specified destination or field set
- Activity: In an authorized isolated test environment, execute TEST-012: Given: certificate lifecycle activities and administrator actions occur When: the system records all administrative interactions with certificates Expected: The system records all administrative interactions with certificates in audit logs

- Activity: In an authorized isolated test environment, execute TEST-013: Given: a system-controlled user opens the dashboard named in the source obligation When: the system provides a centralized dashboard as specified Expected: the requested dashboard is displayed or available for use
- Activity: In an authorized isolated test environment, execute TEST-014: Given: a system-controlled user initiates an exchange with the named external identity service When: the system integrates with existing university identity services as specified Expected: interoperation with that service is observed
- Activity: In an authorized isolated test environment, execute TEST-015: Given: a user initiates use from each stated campus location When: the system supports multiple campus locations as specified Expected: use succeeds at each location
- Activity: In an authorized isolated test environment, execute TEST-016: Given: A draft acceptance plan for this quality goal: the system shall be secure against unauthorized access and data leakage via certificate management operations. When: An analyst inspects or reviews that acceptance plan to determine whether measurable targets are defined or remain TBD. Expected: The plan's measurable targets remain TBD or unconfirmed, as no specific thresholds, attack scenarios, or success criteria are provided in the current draft.
- Activity: In an authorized isolated test environment, execute TEST-017: Given: The system shall remain secure against unauthorized access and data breaches. When: An analyst inspects or reviews that acceptance plan. Expected: The plan's measurable targets remain TBD or unconfirmed.
- Activity: In an authorized isolated test environment, execute TEST-018: Given: The system shall remain reliable under varying loads and failure conditions. When: An analyst inspects or reviews that acceptance plan. Expected: The plan's measurable targets remain TBD or unconfirmed.
- Activity: In an authorized isolated test environment, execute TEST-019: Given: The system shall remain scalable to support increasing numbers of users and data volumes. When: An analyst inspects or reviews that acceptance plan. Expected: The plan's measurable targets remain TBD or unconfirmed.
- Activity: In an authorized isolated test environment, execute TEST-020: Given: The system shall remain manageable through configuration, monitoring, and operational controls. When: An analyst inspects or reviews that acceptance plan. Expected: The plan's measurable targets remain TBD or unconfirmed.
- Activity: Collect stakeholder feedback; agree its cadence with a human reviewer before scheduling cycles.
- Activity: Return accepted stakeholder feedback and observed defects to the backlog and plan the next iteration.
- Deliverable: Requirement-linked implementation increment
- Deliverable: Recorded test evidence
- Deliverable: Reviewed feedback and next-iteration backlog
- Entry criterion: Prior work, source scope and the reviewed test plan are available for iteration planning.
- Exit criterion: A human reviewer has assessed the increment, test evidence and feedback; unresolved changes return to the next iteration.
- Security check: Review implementation and acceptance evidence against FR-011: The system shall generate security alerts.
- Security check: Review implementation and acceptance evidence against NFR-001: The system shall be secure.

- Security check: Review implementation and acceptance evidence against NFR-002: The system shall remain secure.

C.2.3 Release and handoff

Proposed roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Linked requirements: FR-001, FR-002, FR-003, FR-004, FR-005, FR-006, FR-007, FR-008, FR-009, FR-010, FR-011, FR-012, FR-013, FR-014, FR-015, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005

Human gate: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

- Activity: Review the exact requirements baseline, implementation links and verification evidence before a release decision.
- Activity: Carry unresolved source decisions into handoff; generated text is not human approval.
- Deliverable: Requirements and verification evidence package
- Deliverable: Pending human release decision
- Deliverable: Outstanding-decision record
- Entry criterion: Verification evidence and unresolved decisions are available for human review.
- Exit criterion: A human has recorded the release/handoff decision for the reviewed content.
- Security check: Review implementation and acceptance evidence against FR-011: The system shall generate security alerts.
- Security check: Review implementation and acceptance evidence against NFR-001: The system shall be secure.
- Security check: Review implementation and acceptance evidence against NFR-002: The system shall remain secure.

Detailed acceptance criteria, evidence, architecture and the SDLC plan are available in the supporting report and canonical JSON.

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